XRPL Validator — Daily Engineering Report · 2026-09-04

Period: 2026-09-04 00:00 to 24:00 US Eastern (2026-09-04 04:00 to 2026-09-05 04:00 UTC); ledgers #106,748,088 to #106,770,381. Every count from the ledger and the validator's logs stops at #106,770,381. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 46 engine receipts recorded (20 fixed, 0 explained, 26 without a recorded fix).
- **Changes:** 0 deployments, 8 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,748,088–#106,770,381), by step:

Step	Ledgers with a receipt
Step 1 · native engine	46
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,748,884	Step 1 · native engine	2 object(s) differed in content	fixed: commit 0f5126f, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-02	—	#106,748,904	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-03	—	#106,752,895	Step 1 · native engine	1 transaction result(s) differed from rippled's; 7 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 31093f7, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-04	—	#106,753,241	Step 1 · native engine	3 object(s) differed in content	no disposition recorded
R-05	—	#106,753,769	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 21a8437, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-06	—	#106,753,771	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 21a8437, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-07	—	#106,753,953	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-08	—	#106,755,558	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: 623BC401 (deleted), FE469F2F (deleted); 2 object(s) differed in content	fixed: commit bf5936e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-09	—	#106,755,996	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 571A7D65, B0F9535D, 406DABC9; 2 object(s) differed in content	fixed: commit 10b3cf0, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-10	—	#106,755,998	Step 1 · native engine	1 object(s) differed in content	fixed: commit 102021b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-11	—	#106,756,025	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: 2D381F7A, 9F5BA539; 13 object(s) differed in content	no disposition recorded
R-12	—	#106,756,034	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-13	—	#106,756,199	Step 1 · native engine	4 ledger change(s) the network made were absent from ours: 2D381F7A, 9F5BA539, D68289BE; 3 object(s) differed in content	no disposition recorded
R-14	—	#106,756,521	Step 1 · native engine	1 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-15	—	#106,757,292	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-16	—	#106,757,502	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-17	—	#106,758,110	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 23C3D7F7, 6D3ABEBD, 4E8645C0; 3 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: commit 6283d40, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-18	—	#106,758,117	Step 1 · native engine	8 object(s) differed in content	no disposition recorded
R-19	—	#106,758,148	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: 2DE7C6A0; 6 object(s) differed in content	fixed: commit 962e81b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-20	—	#106,758,166	Step 1 · native engine	2 change(s) of ours the network did not make; 2 object(s) differed in content	no disposition recorded
R-21	—	#106,758,320	Step 1 · native engine	1 object(s) differed in content	fixed: commit c1c1980, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-22	—	#106,758,324	Step 1 · native engine	1 object(s) differed in content	fixed: commit fd8379b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-23	—	#106,758,384	Step 1 · native engine	1 object(s) differed in content	fixed: commit fd8379b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-24	—	#106,758,514	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: E97A2ADC; 3 object(s) differed in content	no disposition recorded
R-25	—	#106,759,265	Step 1 · native engine	2 transaction result(s) differed from rippled's; 2 ledger change(s) the network made were absent from ours: 8F941C06, C40FCFD0; 5 object(s) differed in content	fixed: commit 51a6221, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-26	—	#106,759,499	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: DE5F9594 (deleted), 39528E5C (deleted); 10 object(s) differed in content	fixed: commit 059b37e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-27	—	#106,759,500	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit f314dca, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-28	—	#106,759,512	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-29	—	#106,759,539	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-30	—	#106,759,810	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-31	—	#106,759,872	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-32	—	#106,759,960	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-33	—	#106,761,226	Step 1 · native engine	1 object(s) differed in content	fixed: commit 48928e9, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-34	—	#106,762,519	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-35	—	#106,763,015	Step 1 · native engine	1 object(s) differed in content	fixed: commit fab8f1e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-36	—	#106,763,704	Step 1 · native engine	1 transaction result(s) differed from rippled's; 7 change(s) of ours the network did not make	no disposition recorded
R-37	—	#106,764,402	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 3366C527, 8817D64F, 578C5D14; 2 object(s) differed in content	no disposition recorded
R-38	—	#106,765,989	Step 1 · native engine	1 transaction result(s) differed from rippled's; 7 change(s) of ours the network did not make	no disposition recorded
R-39	—	#106,766,906	Step 1 · native engine	3 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: commit 128b564, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-40	—	#106,766,924	Step 1 · native engine	2 ledger change(s) the network made were absent from ours: EFAD644D, 036F3B98; 5 object(s) differed in content	fixed: commit d00a396, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-04-01, INC-01 is INC-2026-09-04-01. Times are not shown where the reference node no longer holds the ledger.

6 more receipts (#106,767,718–#106,769,882) are in the record and not listed individually; their dispositions: 1 fixed, 5 no disposition recorded.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
04:33 UTC	xrpl-validator	`8405e57`	fix(ledger): a stranger may burn an NFToken only if it is burnable and they are its issuer or minter (finding 154)

Merged	Repo	Commit	Subject
05:17 UTC	xrpl-validator	`0f5126f`	fix(ledger): a two-asset AMM deposit mints its sized tokens, LPTokenOut is only a floor (finding 155)
09:17 UTC	xrpl-validator	`31093f7`	fix(ledger): an AMM withdrawal charges reserve for a trust line it must create (finding 156)
10:29 UTC	xrpl-validator	`21a8437`	fix(ledger): a direct hop into a globally frozen account is dry (finding 157)
12:30 UTC	xrpl-validator	`bf5936e`	fix(ledger): a self-cross removes every self-offer inside the limit, not just the tip (finding 158)
13:29 UTC	xrpl-validator	`10b3cf0`	fix(ledger): a crossing pass is judged on its pool slice plus its fills (finding 159)
13:29 UTC	xrpl-validator	`102021b`	fix(ledger): a tfSell residual divides the gross by the rate with rippled's lossy divRound (finding 160)
21:15 UTC	xrpl-validator	`4e7e5b4`	fix(node): RPC client reports the real count and time spent on a fallback endpoint

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): a stranger may burn an NFToken only if it is burnable and they are its issuer or minter (finding 154)** (`8405e57`, xrpl-validator) NFTokenBurn::preclaim lets a token's owner burn it unconditionally; any other account needs the token flagged burnable (0x0001 in the NFTokenID's flag word) and must be its issuer or the issuer's current NFTokenMinter — an issuer with no minter set fails that comparison, an issuer whose root is gone skips it (NFTokenBurn.cpp preclaim). We had no permission check at all: every burn with an Owner field went through.
- **fix(ledger): a two-asset AMM deposit mints its sized tokens, LPTokenOut is only a floor (finding 155)** (`0f5126f`, xrpl-validator) On the tfTwoAsset path rippled sizes the deposit from Amount/Amount2 and the pool's balances — `equalDepositLimit` picks the leg whose proportional counterpart fits, rounds the tokens (`getRoundedLPTokens`) and mints exactly those — and treats an LPTokenOut on the transaction only as `lpTokensDepositMin`: `deposit()` fails with tecAMM_FAILED when the sized tokens fall below it, before anything moves. We minted the …
- **fix(ledger): an AMM withdrawal charges reserve for a trust line it must create (finding 156)** (`31093f7`, xrpl-validator) `AMMWithdraw::withdraw` runs `sufficientReserve` for every asset it delivers (fixAMMv1_2): when the account holds no trust line for an IOU being withdrawn, the line it is about to create is a new owned object, and TrustSet's rule applies — nothing owed while the account owns fewer than two objects, otherwise base plus increment times OwnerCount + 1, measured against the balance before the fee (`max(priorBalance, …
- **fix(ledger): a direct hop into a globally frozen account is dry (finding 157)** (`21a8437`, xrpl-validator) `checkFreeze` (StepChecks.h:19-46) runs on every DirectStep that is not the strand's only step: the hop is terNO_LINE when the DESTINATION account carries lsfGlobalFreeze — the holder's own flag, whatever the issuer — when the (src, dst) trust line carries the destination's freeze bit, or when either side deep-froze it. Our freeze model looked only at the issuer's global freeze and the issuer-side line flag, so a …
- **fix(ledger): a self-cross removes every self-offer inside the limit, not just the tip (finding 158)** (`bf5936e`, xrpl-validator) Ledger #106755558, tx 15B9744CE01E: rw7nJtEN places an offer selling 256225 BBRL for

50000 RLUSD (limit 5.1243 BBRL per RLUSD). The direct tip is the account's own 20000 RLUSD bid at 5.1232 and the next level its own bid at 5.1238, both inside the limit. rippled's execOffer (BookStep.cpp:439-451) removes the first through limitSelfCrossQuality and, since no offer has been attempted yet, resets the level quality …

- **fix(ledger): a crossing pass is judged on its pool slice plus its fills (finding 159)** (`10b3cf0` , xrpl-validator) Ledger #106755996, tx D84E769E7583: rhhh49pF's tfImmediateOrCancel offer sells 21.88 XRP for 210.047995624 CNY. The direct book's tip (rwvLbHQt, 210.048 CNY for 21880000 drops) prices exactly at the taker's limit, and the XRP/CNY pool is anchored on it: tryAMM takes 6980735 drops for 67.015056 CNY, a hair inside the limit, then execOffer fills the tip for the remaining 143.032939624 CNY at the drop ceiling, 14899265 …
- **fix(ledger): a tfSell residual divides the gross by the rate with rippled's lossy divRound (finding 160)** (`102021b` , xrpl-validator) Ledger #106755998, tx A3D3541FD33F: rURtT5MM's tfSell offer sells 0.12393 BTC (issuer rate 1.0015) for RLUSD and crosses one offer, debiting 0.05880101336798101 BTC gross. OfferCreate.cpp derives the rested TakerGets as takerGets minus divideRound(actualAmountIn, gatewayXferRate, asset, roundUp), and divideRound is STAmount's divRound: muldivRound ceils the 17-18 digit quotient, then canonicalizeRound truncates an …
- **fix(node): RPC client reports the real count and time spent on a fallback endpoint** (`4e7e5b4` , xrpl-validator) The periodic 'Trying primary endpoint again after N successes on fallback' line printed a counter that was never reset, so N was the process's lifetime success count (1,816,400 this afternoon) and read as hours parked on s1.ripple.com when the client had in fact been back on the the reference node primary within a second of every failover.

Independence program

No independence record for UTC day 2026-09-04; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

- Receipt R-02 (#106,748,904): open, under investigation.
- Receipt R-04 (#106,753,241): open, under investigation.
- Receipt R-07 (#106,753,953): open, under investigation.
- Receipt R-11 (#106,756,025): open, under investigation.
- Receipt R-12 (#106,756,034): open, under investigation.

- Receipt R-13 (#106,756,199): open, under investigation.
- Receipt R-14 (#106,756,521): open, under investigation.
- Receipt R-15 (#106,757,292): open, under investigation.
- Receipt R-16 (#106,757,502): open, under investigation.
- Receipt R-18 (#106,758,117): open, under investigation.
- Receipt R-20 (#106,758,166): open, under investigation.
- Receipt R-24 (#106,758,514): open, under investigation.
- Receipt R-28 (#106,759,512): open, under investigation.
- Receipt R-29 (#106,759,539): open, under investigation.
- Receipt R-30 (#106,759,810): open, under investigation.
- Receipt R-31 (#106,759,872): open, under investigation.
- Receipt R-32 (#106,759,960): open, under investigation.
- Receipt R-34 (#106,762,519): open, under investigation.
- Receipt R-36 (#106,763,704): open, under investigation.
- Receipt R-37 (#106,764,402): open, under investigation.
- Receipt R-38 (#106,765,989): open, under investigation.
- Receipt R-41 (#106,767,718): open, under investigation.
- Receipt R-42 (#106,768,367): open, under investigation.
- Receipt R-43 (#106,769,252): open, under investigation.
- Receipt R-44 (#106,769,269): open, under investigation.
- Receipt R-46 (#106,769,882): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **AMM Price-Triggered Stop Offers** (dex) Add a tfTriggerOnAMM flag plus sfTriggerAMMID (AMM keylet), sfTriggerDirection, and sfTriggerPrice (STAmount) SFields to OfferCreate. A triggered offe
- **On-Ledger Travel Rule Envelope for Payments** (compliance) Add an optional sfTravelRuleEnvelope VL SField (opaque ciphertext, capped at 4KB) to Payment, plus a new AccountRoot flag lsfRequireTravelRule and pai
- **Hash-Blind Offer Tiebreak for MEV Resistance** (dex) Change the tie-break rule for equal-quality Offers landing in the same ledger's SortedDirectories-ordered book: instead of ordering ties by canonical

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-04 03:54	09-04 04:33	xrpl-validator	`8405e57`	fix(ledger): a stranger may burn an NFToken only if it is burnable and they are its issuer or minter (finding 154)
09-04 05:00	09-04 05:17	xrpl-validator	`0f5126f`	fix(ledger): a two-asset AMM deposit mints its sized tokens, LPTokenOut is only a floor (finding 155)
09-04 09:17	09-04 09:17	xrpl-validator	`31093f7`	fix(ledger): an AMM withdrawal charges reserve for a trust line it must create (finding 156)
09-04 10:29	09-04 10:29	xrpl-validator	`21a8437`	fix(ledger): a direct hop into a globally frozen account is dry (finding 157)
09-04 12:30	09-04 12:30	xrpl-validator	`bf5936e`	fix(ledger): a self-cross removes every self-offer inside the limit, not just the tip (finding 158)
09-04 12:57	09-04 13:29	xrpl-validator	`10b3cf0`	fix(ledger): a crossing pass is judged on its pool slice plus its fills (finding 159)
09-04 12:57	09-04 13:29	xrpl-validator	`102021b`	fix(ledger): a tfSell residual divides the gross by the rate with rippled's lossy divRound (finding 160)
09-04 21:15	09-04 21:15	xrpl-validator	`4e7e5b4`	fix(node): RPC client reports the real count and time spent on a fallback endpoint

Times in UTC. A commit appears on the day it was committed to main or merged into it.